

CSC311 Introduction to Machine Learning

Bias Variance Decomposition, Ensemble Methods

Slides courtesy of Alice Gao

Learning Outcomes

By the end of this lecture, you should be able to

- Compare multiple models in terms of bias/variance.
- Identify the sources of errors for a given model in terms of bias, variance, and Bayes error.
- Explain the key advantages of bagging.
- Describe the process of creating a bagged model for binary classification.

Outline

- Define Bias and Variance
- Mathematical Interpretation
- Bagging

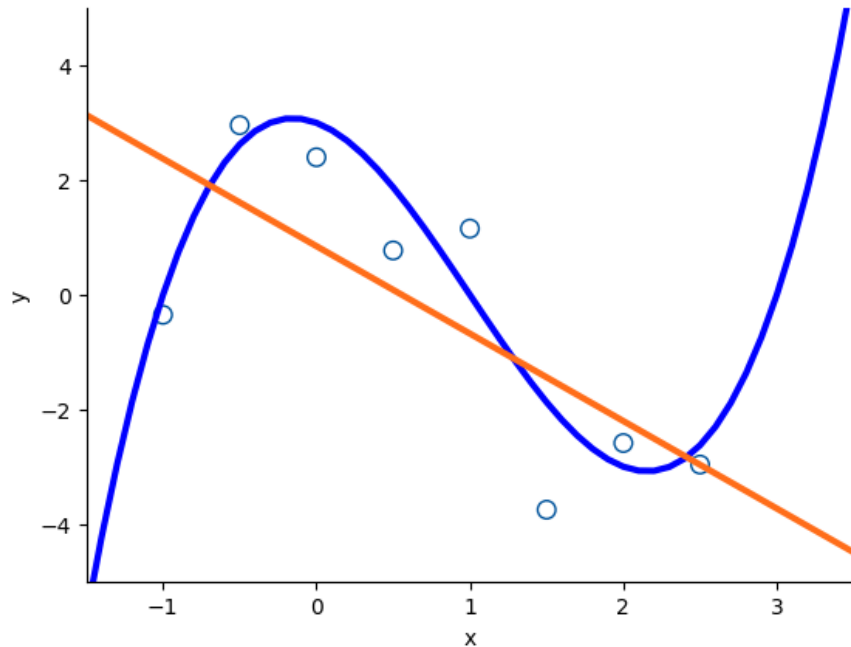


Defining Bias and Variance

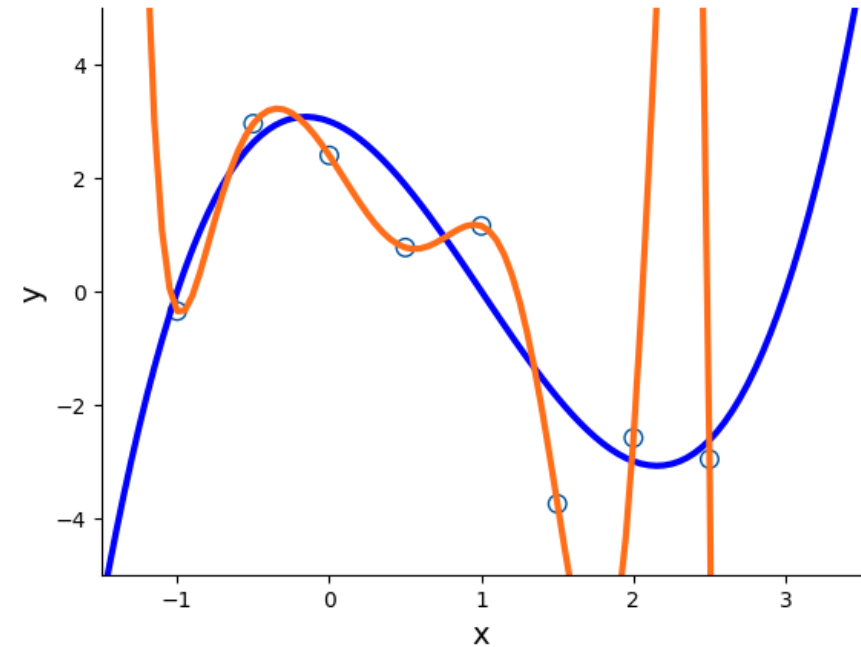
Sources of Errors in Different Models

Errors in the two figures differ significantly!

Underfitting



Overfitting



Let's systematically analyze where errors come from.

Summary: Bias Variance Decomposition

Generalization Error =

Bias	+	Variance	+	Bayes Error
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Model's inability to
capture true relationship

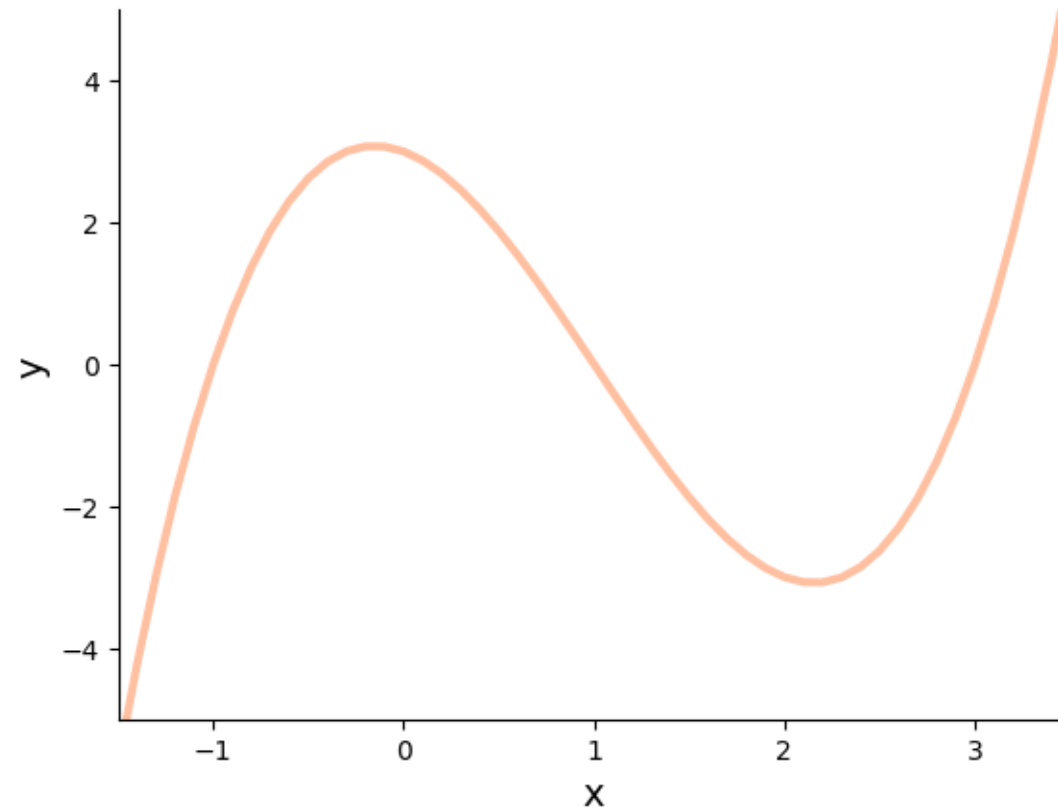
$\approx$ underfitting

Model's sensitivity to
random fluctuations

$\approx$ overfitting

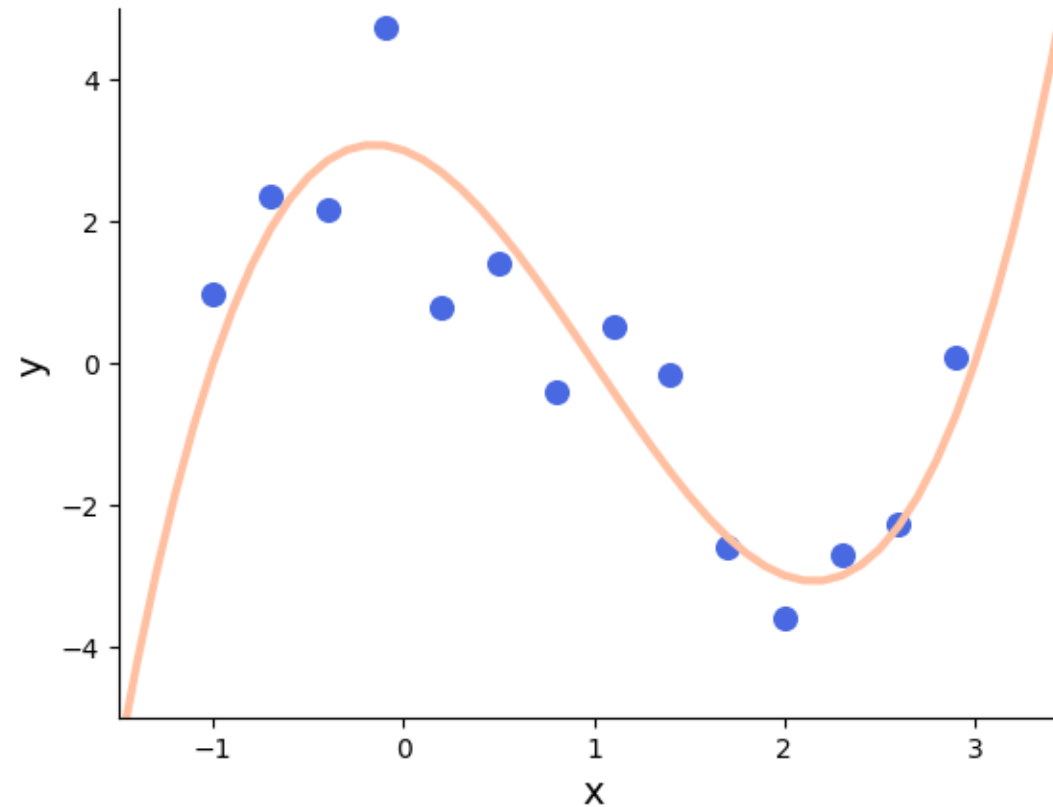
Inherent noise
in the problem

The Data Generating Distribution



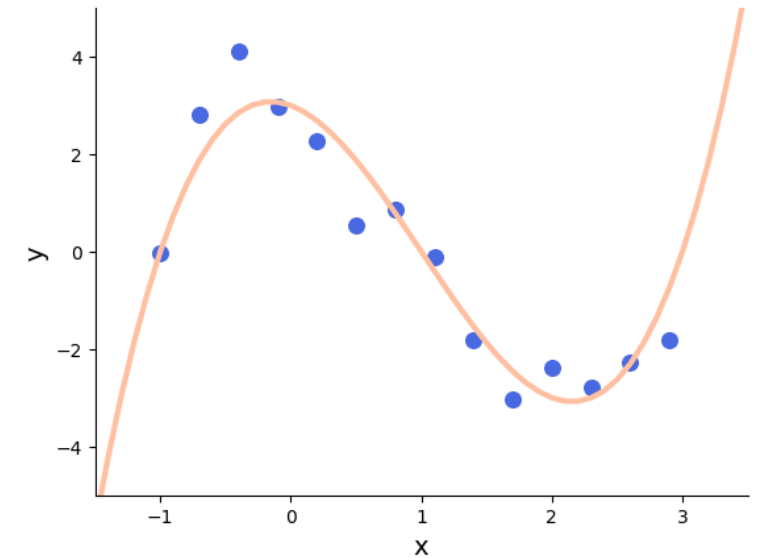
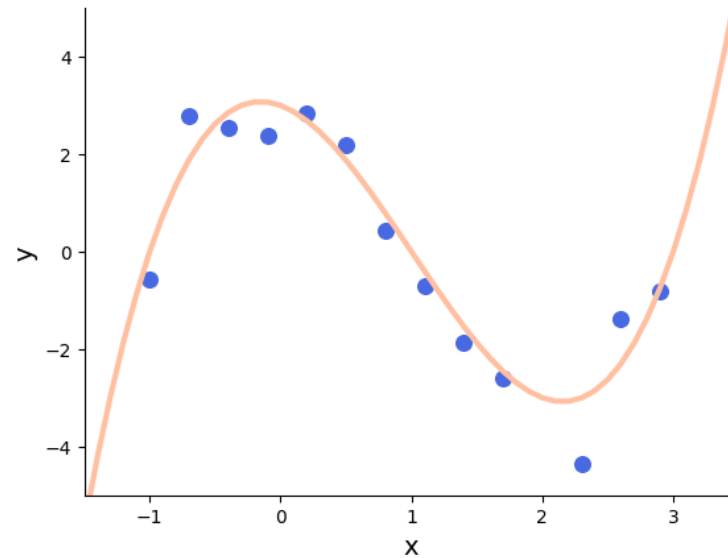
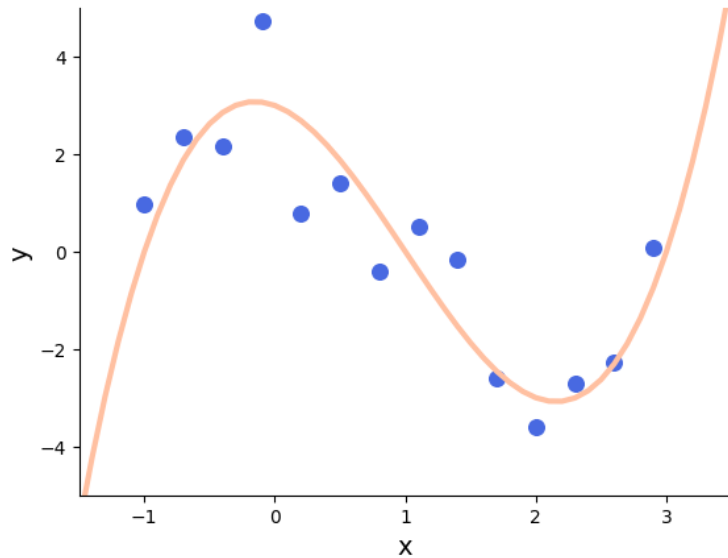
Sampling a Training Set

Sample from the data generating distribution with unbiased noise



Sampling Different Training Sets

We can sample different training sets from the data generating distribution.



Comparing Two Models

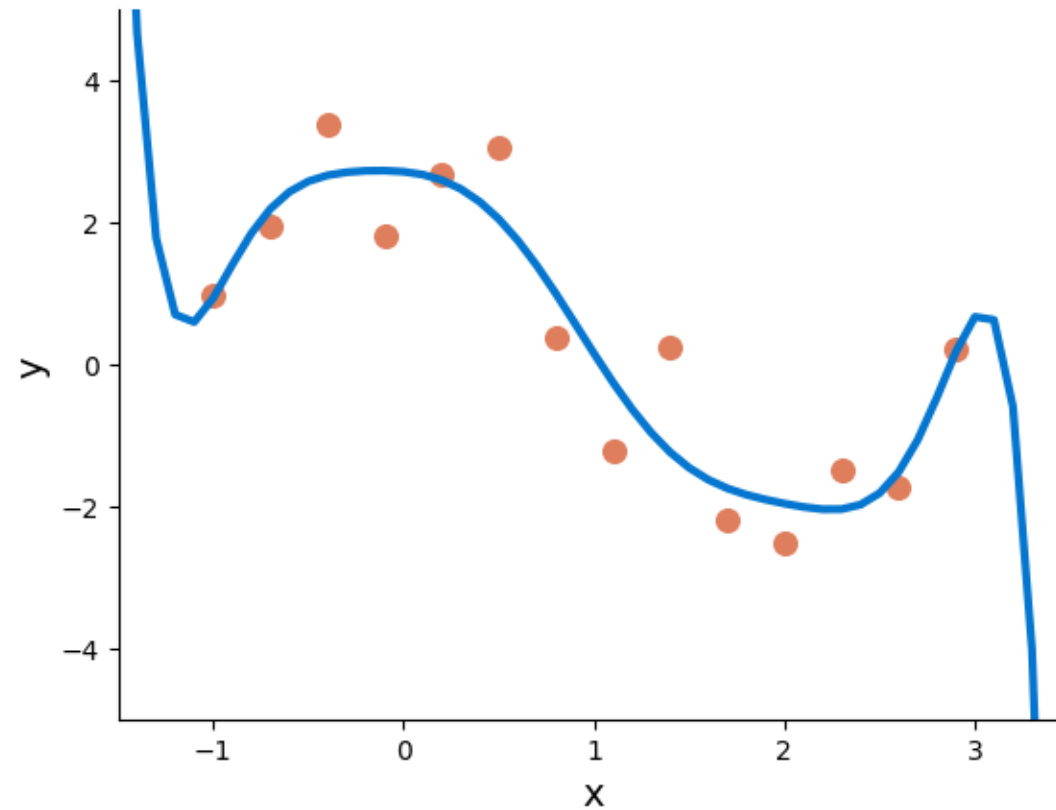
Two models:

- (Polynomial Regression) Fitting a high-degree polynomial
- (Linear Regression) Fitting a line

Variance: How much do the **predictions vary** for different training sets?

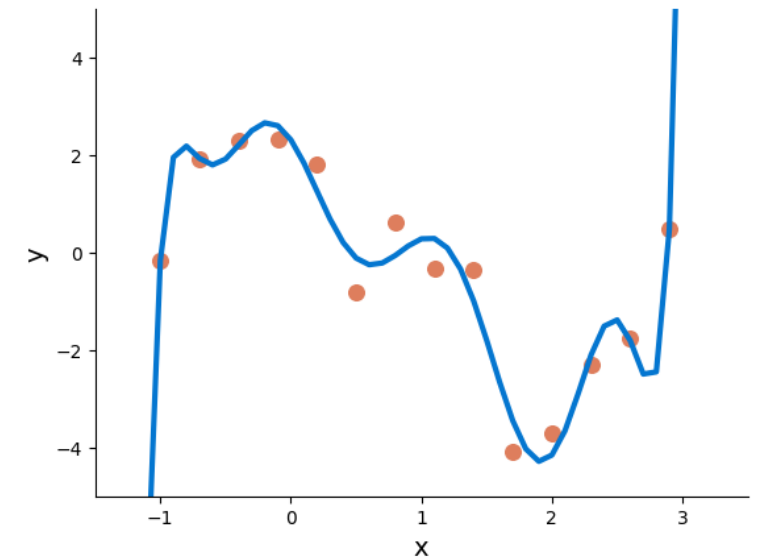
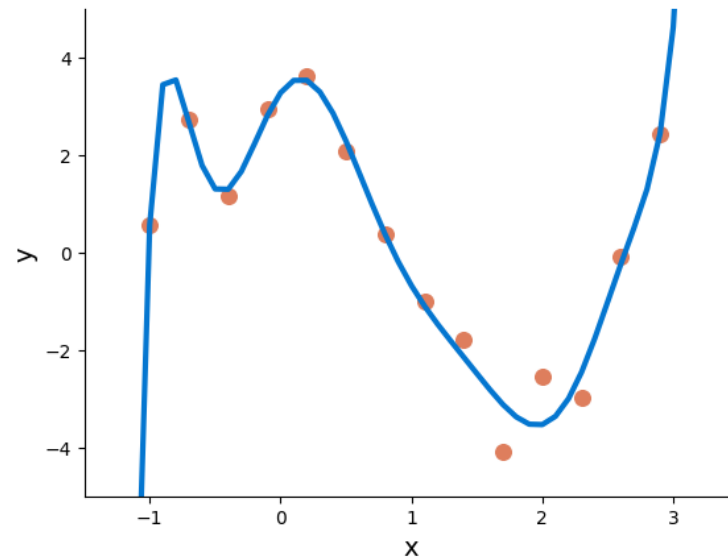
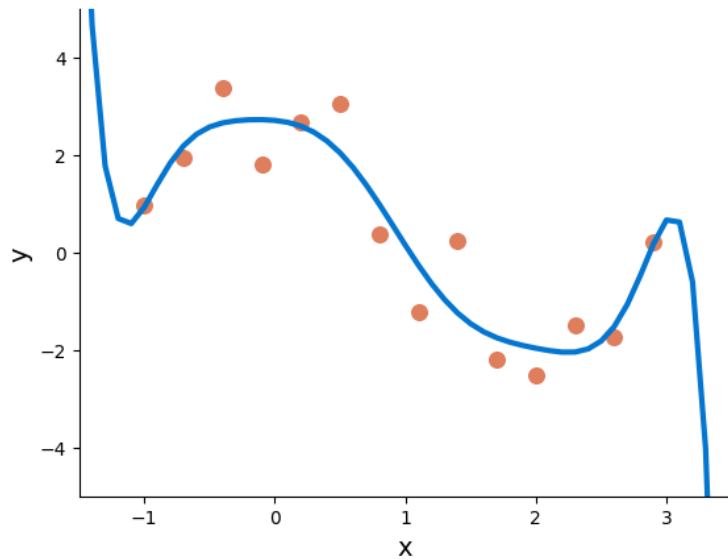
Bias: How **wrong** is the **average prediction** compared to the expected target?

Fitting a High-Degree Polynomial

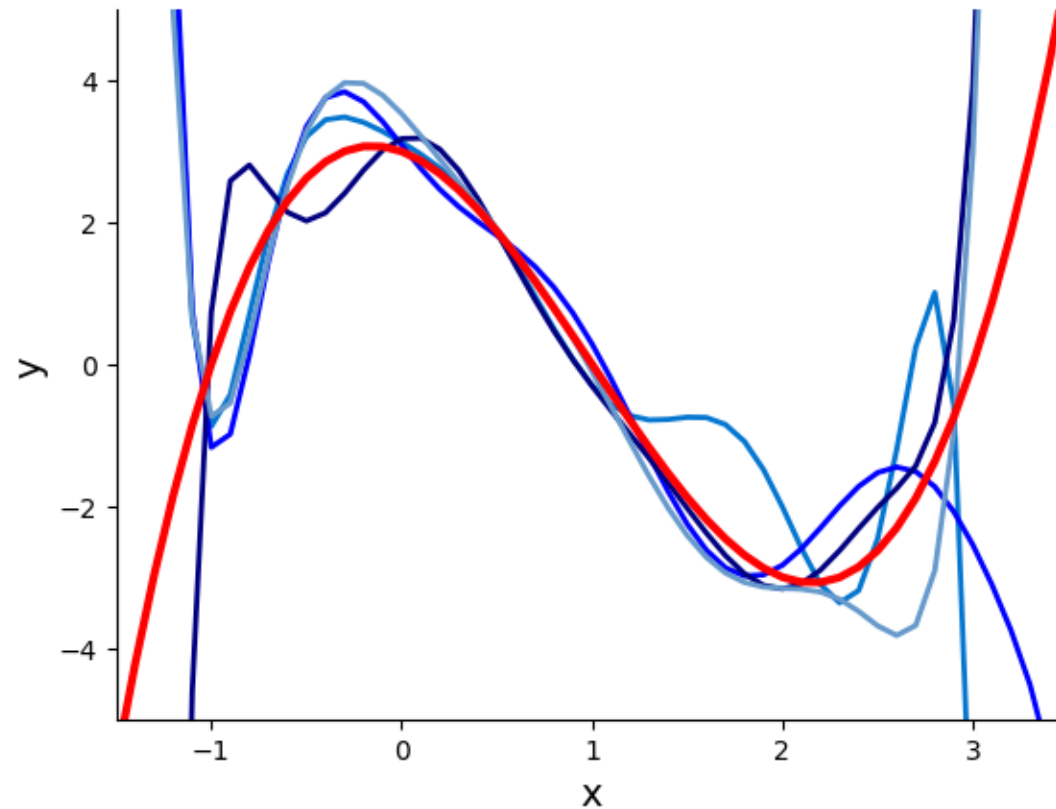


Fitting Polynomials for Different Training Sets

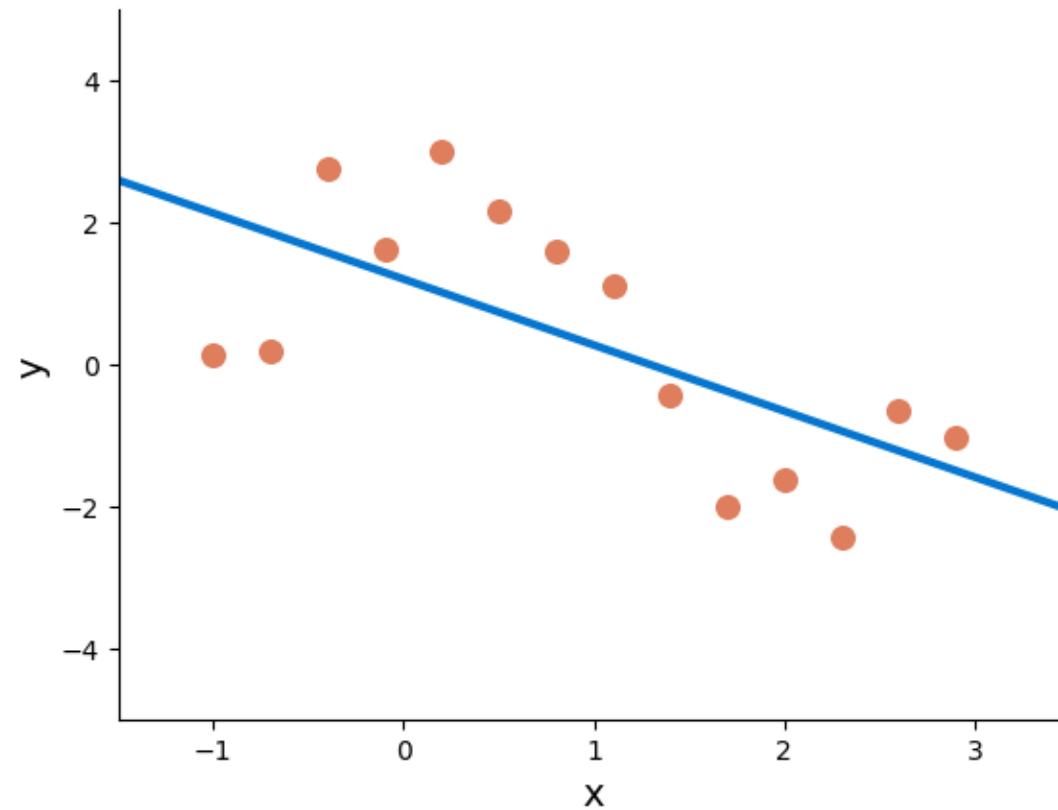
We get a different model each time we sample a new training set.



Comparing Polynomials for Different Training Sets

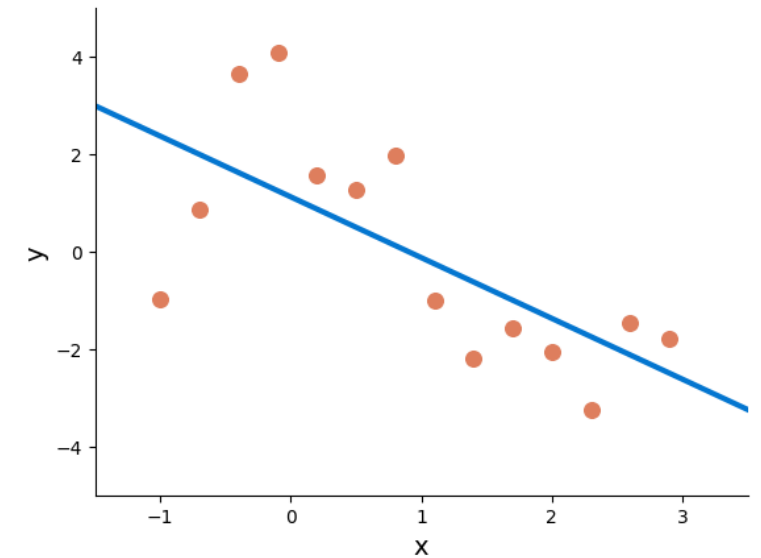
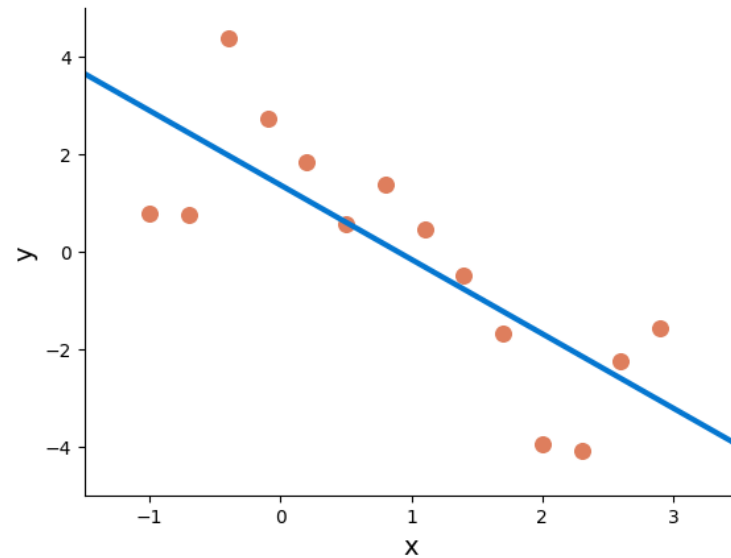
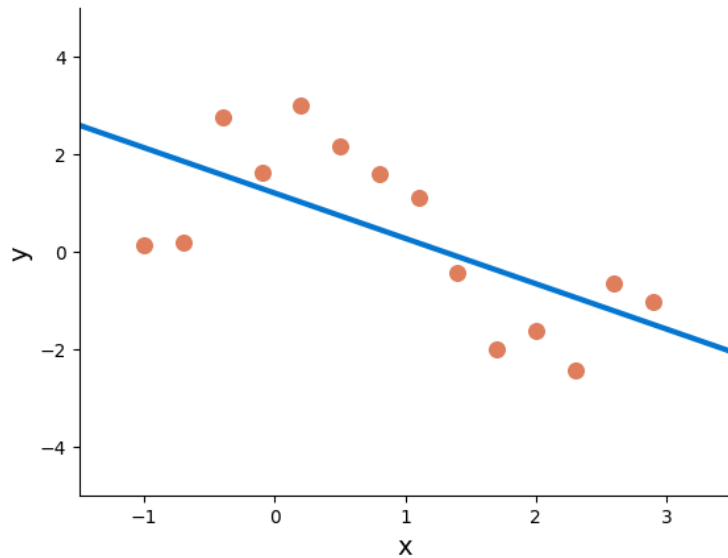


Fitting a Line

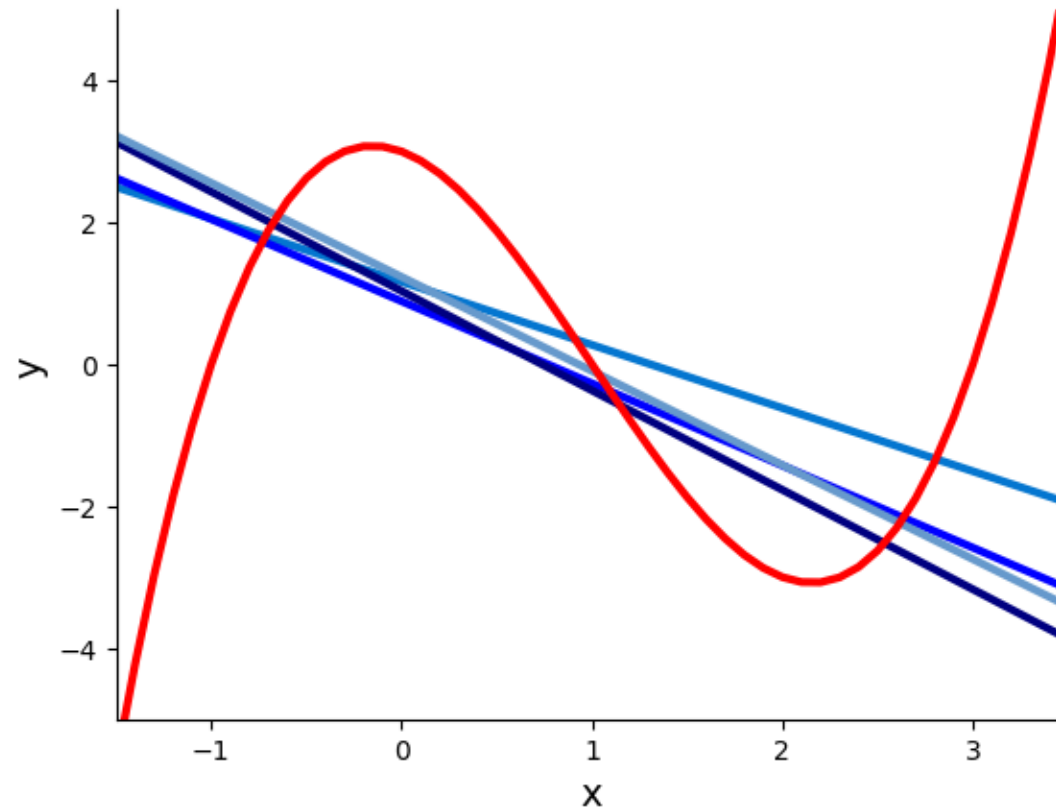


Fitting Lines for Different Training Sets

We get a different model each time we sample a new training set.



Comparing Lines for Different Training Sets

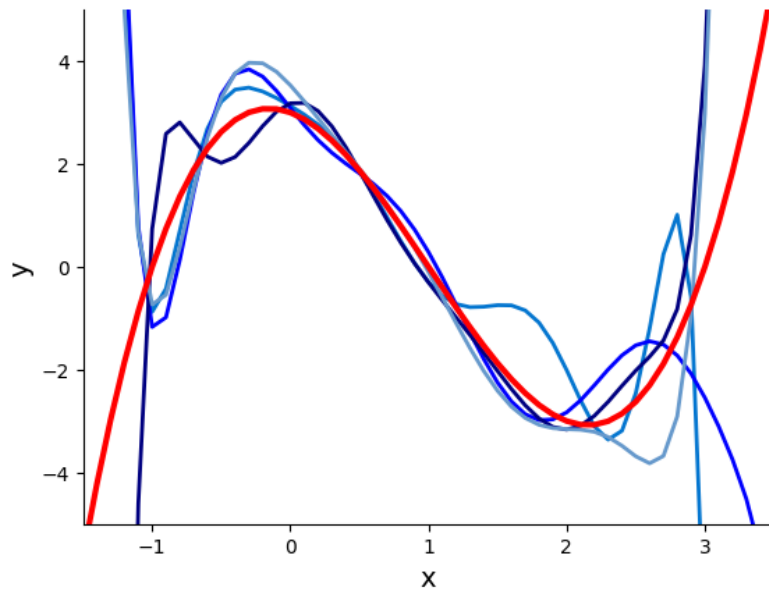


Variance of the Two Models

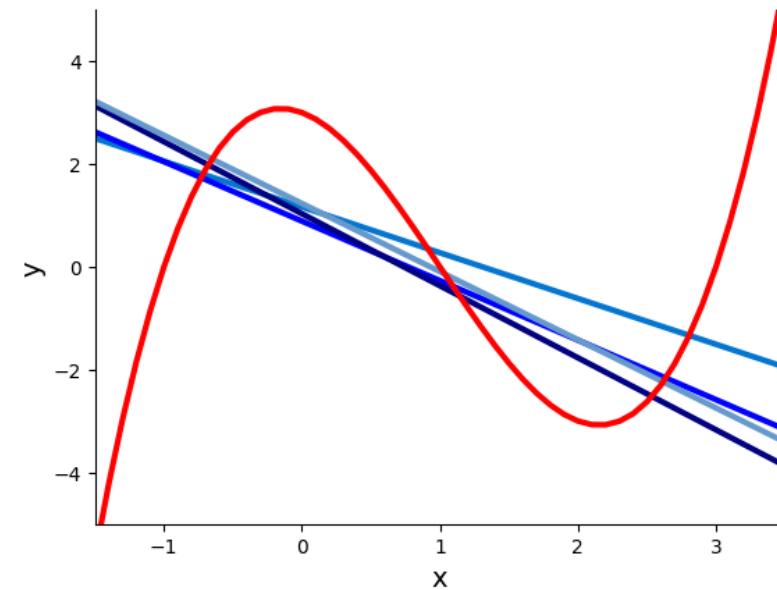
Variance measures how much the predictions vary for different training sets.

Which model has a **larger variance**?

(A) Polynomial



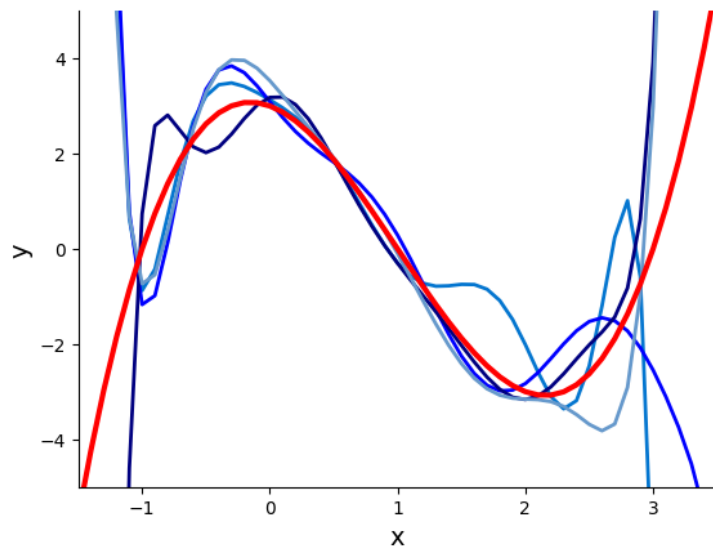
(B) Line



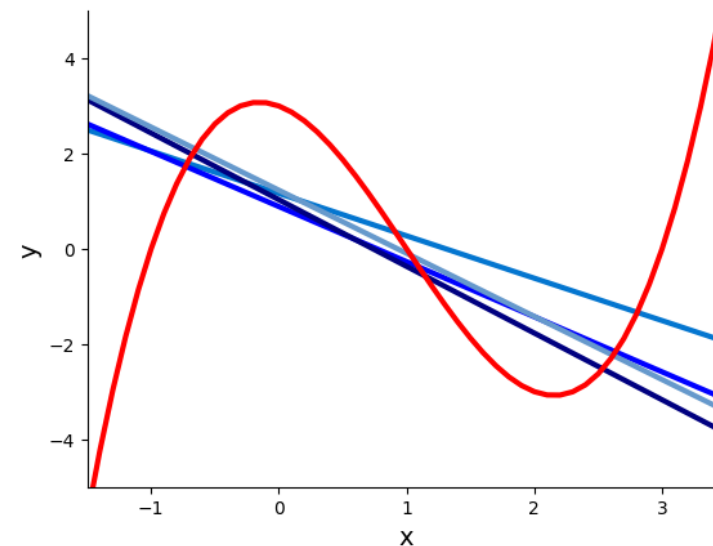
Bias of the Two Models

Bias measures how wrong average prediction is compared to the expected target. Which model has a **larger bias**?

(A) Polynomial



(B) Line

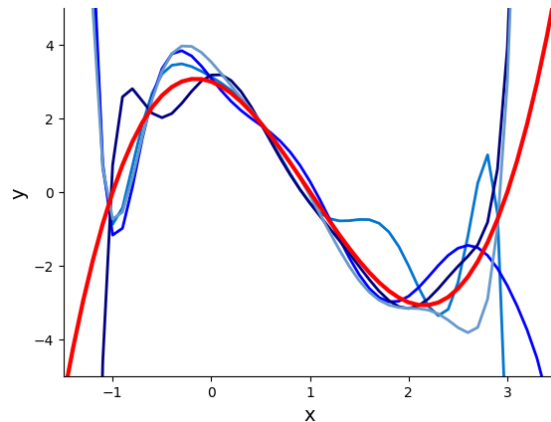


Polynomial versus Linear Regression

Polynomial Regression

High Variance: Small changes in the training data results in large changes in the predictions.

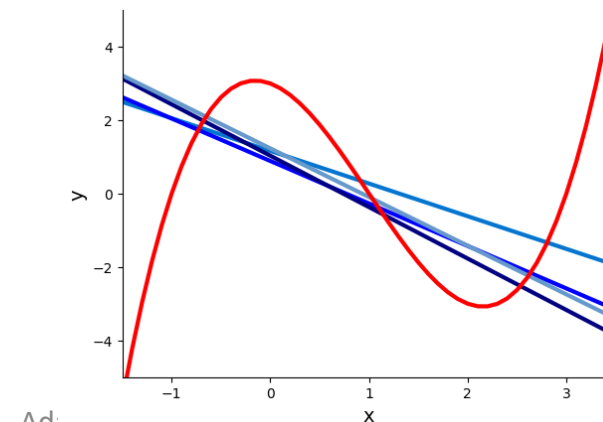
Low Bias: Average prediction across models is about right.



Linear Regression

Low Variance: Small changes in the training data result in little change in the predictions.

High Bias: Predictions are systematically more wrong than a good model.





A Mathematical Interpretation

A Mathematical Interpretation

- $p_{\mathcal{D}}$ denotes the data generating distribution.
- Sample a training set $\mathcal{D}$ from $p_{\mathcal{D}}$.
- A learning algorithm $\mathcal{A}$ takes the training set $\mathcal{D}$ as input and produces a function $f: \mathbf{x} \rightarrow y$ as output.
- Sample a test example $(\mathbf{x}, t)$ from $p_{\mathcal{D}}$ and make a prediction $y = f(\mathbf{x})$.
- Compute test error using squared loss: $\mathcal{L}(y, t) = (y - t)^2$.
- We will analyze the expected test error

$$\mathbb{E}[(y - t)^2]$$

Expectation with Respect to What?

$$\mathbb{E}[(y - t)^2]$$

What is the expectation with respect to?

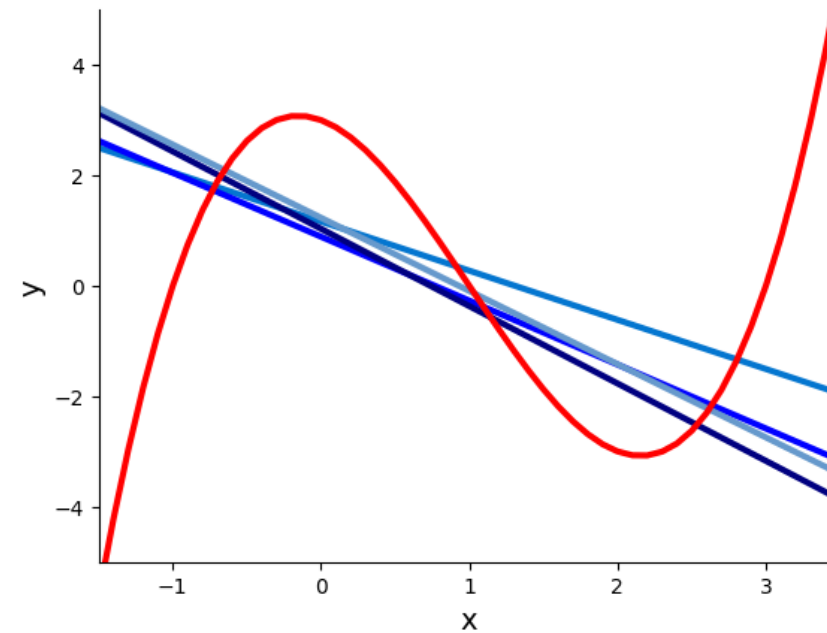
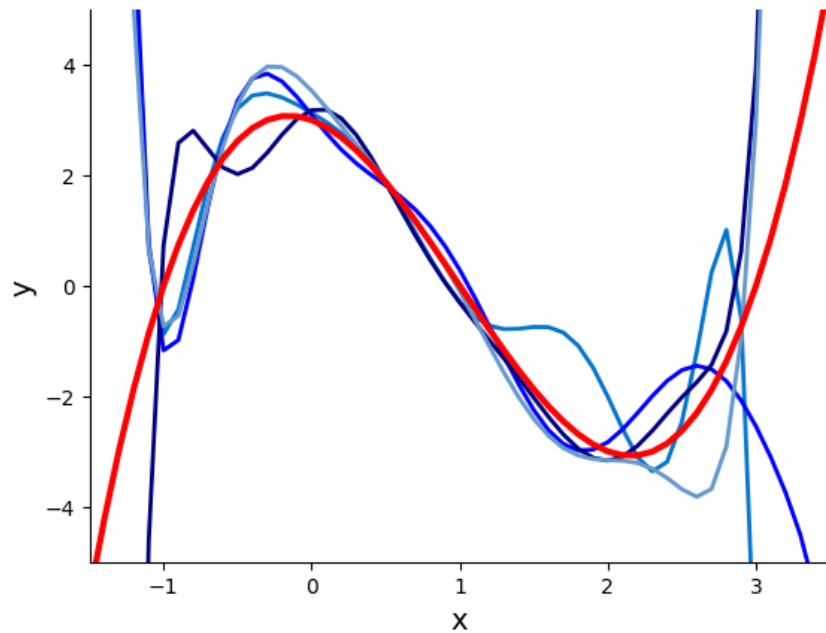
- Training set $\mathcal{D}$ randomly sampled from data generating distribution $p_{\mathcal{D}}$.
- Test point $(\mathbf{x}, t)$ randomly sampled from data generating distribution $p_{\mathcal{D}}$.

$$\mathbb{E}_{\mathbf{x}, t, \mathcal{D}}[(y - t)^2]$$

The Average Predictor $\bar{y}$

$$\bar{y} = \mathbb{E}_{\mathcal{D}}[y]$$

the **average** of predictors trained on **all possible training sets** drawn from the same data generating distribution



Decomposing Expected Test Error: Step 1

$$\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - t)^2]$$

Subtract and add the average predictor $\bar{y}$.

$$\begin{aligned}\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - t)^2] &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - \bar{y} + \bar{y} - t)^2] \\ &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - \bar{y})^2] + 2\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - \bar{y})(\bar{y} - t)] + \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - t)^2] \\ &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - \bar{y})^2] + 0 + \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - t)^2]\end{aligned}$$

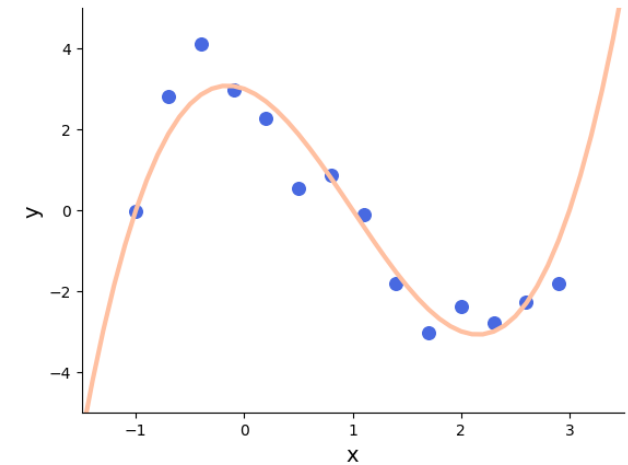
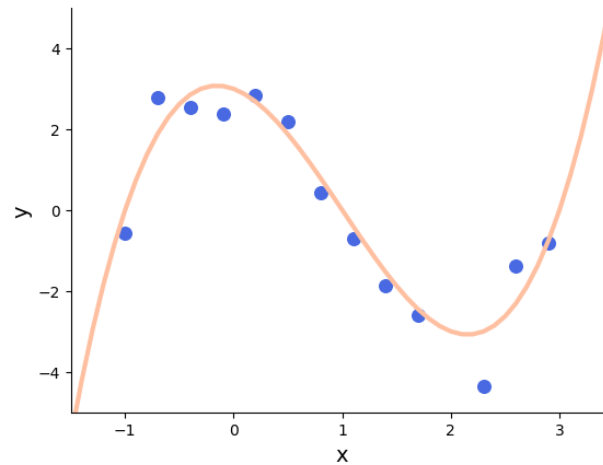
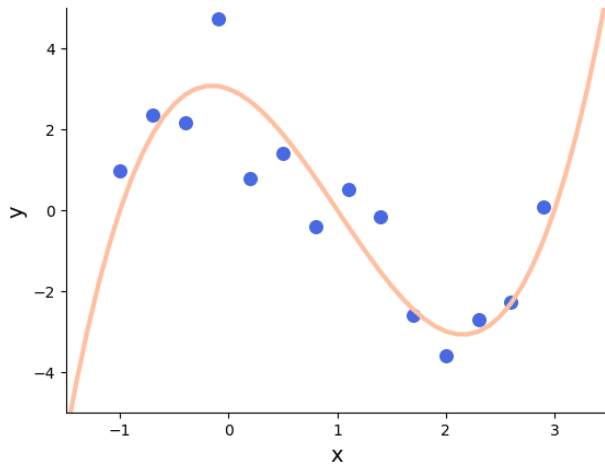
(To see why the middle term is zero, try pushing just $\mathbb{E}_{\mathcal{D}}$ into the bracket first.)

The Optimal Predictor y^*

$$y^* = \mathbb{E}_t[t|\mathbf{x}]$$

the **expected target** value at input $\mathbf{x}$

the best possible prediction for input $\mathbf{x}$ given the noisy data generating process



Decomposing Expected Test Error: Step 2

$$\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - t)^2] = \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - \bar{y})^2] + \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - t)^2]$$

Let's further decompose $\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - t)^2]$ by subtracting and adding y^* .

$$\begin{aligned}\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - t)^2] &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - y^* + y^* - t)^2] \\ &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - y^*)^2] + 2\mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - y^*)(y^* - t)] + \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y^* - t)^2] \\ &= \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(\bar{y} - y^*)^2] + \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y^* - t)^2]\end{aligned}$$

Summarizing the Decomposition

$$\begin{aligned} & \mathbb{E}_{\mathbf{x},t,\mathcal{D}}[(y - t)^2] \\ &= \mathbb{E}_{\mathbf{x},\mathcal{D}}[(y - \bar{y})^2] \qquad \qquad \qquad + \mathbb{E}_{\mathbf{x}}[(\bar{y} - y^*)^2] \qquad \qquad \qquad + \mathbb{E}_{\mathbf{x},t}[(y^* - t)^2] \end{aligned}$$

Variance:

Expected diff between prediction y and average prediction $\bar{y}$. Can be expressed $\text{Var}(y)$.

How much the predictions vary for different training sets

Bias²:

Expected squared diff between average prediction $\bar{y}$ and expected target y^*

How wrong is the average prediction compared to expected target

Bayes Error:

Noise inherent in the problem

Interpreting Variance, Bias, and Bayes Error

Variance

Appears as **large variation of predictions** across training sets.

Arises from algorithm **being sensitive to noise** in the training set.

Overfitting

Bias

Appears as the **average prediction** being **systematically wrong**.

Arises from the algorithm's **erroneous assumption** about the data.

Underfitting

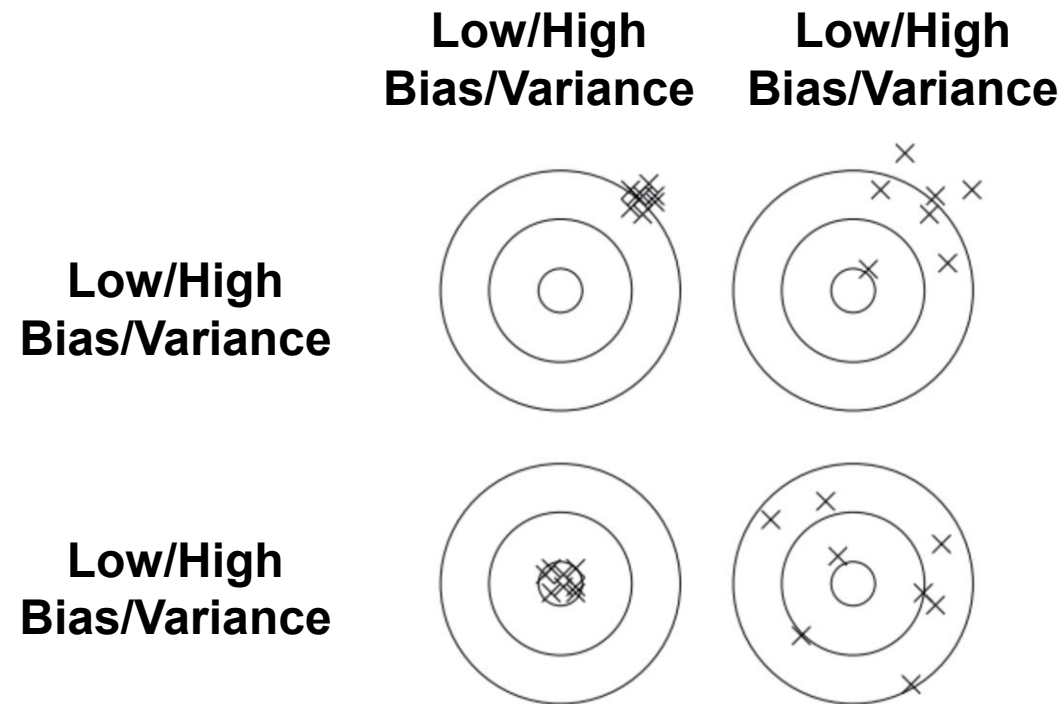
Bayes Error

Appears as **inherent noise** in the data. Prevent any model from achieving zero loss.

Arises from multiple **test points** having the **same features** but **different targets**.

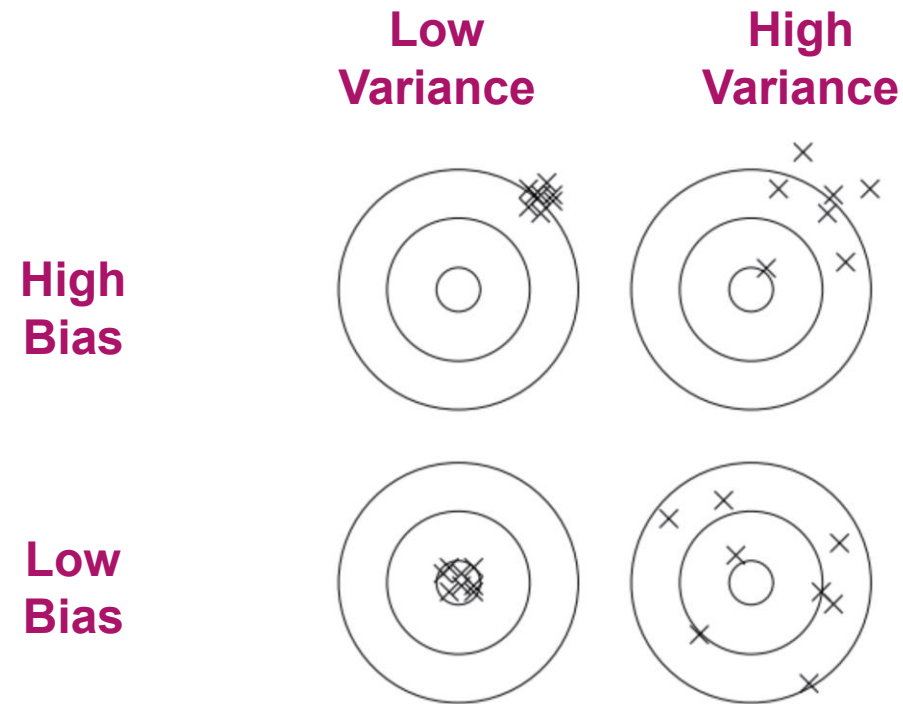
Bias Variance Dartboard

Label each dartboard with low/high bias/variance.



Solution: Bias Variance Dartboard

Label each dartboard with low/high bias/variance.



Example of Bayes Error

Multiple test data points have same features $\mathbf{x}$ but different targets t .

Is Bayes error irreducible?

- Yes, given current features and model.
- Can be reduced by creating new features that reveal previously hidden patterns in data.

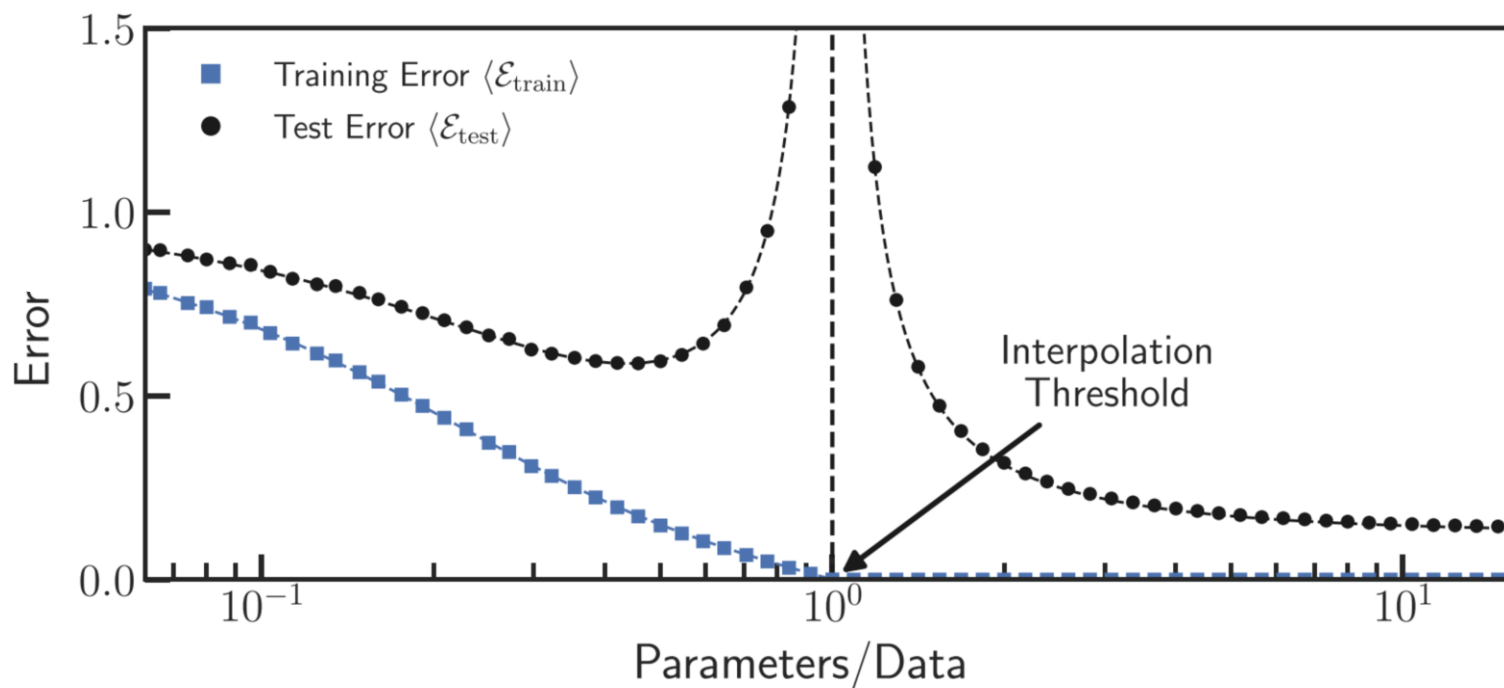
Data Point	A	B	Label
1	large	10	yes
2	large	20	no
3	large	20	no
4	large	20	no
5	medium	10	yes
6	medium	10	yes
7	medium	10	no
8	medium	20	no
9	small	10	yes
10	small	20	yes
11	small	20	no

Table 2: Data Set for Decision Trees

Do low bias models always have high variance?

As it turns out, not always!

Sometimes, when a model has so many parameters that they outnumber training examples, we see **double descent**.



Do low bias models always have high variance?

This can occur because more parameters than examples means there are multiple possible solutions, and many of those solutions might be pretty good!

This is particularly important in deep learning, where models can have an enormous number of parameters. We don't always observe double descent in practice, though.

For a high-level explanation, see:

<https://threadreaderapp.com/thread/1292293102103748609.html>



Ensemble Methods: Bagging

Ensemble Methods: Stronger Together

“The wisdom of crowds often outperforms individual brilliance.”

Combine predictions from multiple models

for **lower variance**, better generalization, and greater robustness.

Popular techniques:

- Bagging: Train on random subsamples and combine results.
- Boosting: Train sequentially, correcting past errors.

Bagging: Motivation

Suppose we could

- sample m independent training sets from p_D
- compute the prediction y_i for each training set
- average the predictions

$$y = \frac{1}{m} \sum_{i=1}^m y_i$$

How does averaging the predictions affect
the bias, the variance, and Bayes error?

Bagging: Why It Works

How does averaging the predictions affect the bias, variance and Bayes error?

$$y = \frac{1}{m} \sum_{i=1}^m y_i$$

- Bayes error: increased / unchanged / decreased
- Bias: increased / unchanged / decreased

$$\bar{y} = \mathbb{E}[y] = \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m y_i \right] = \underline{\hspace{2cm}}$$

- Variance: increased / unchanged / decreased

$$\text{Var}[y] = \text{Var} \left[\frac{1}{m} \sum_{i=1}^m y_i \right] = \underline{\hspace{2cm}}$$

(Remember, $\text{Var}[y] = \mathbb{E}_{\mathbf{x}, \mathcal{D}}[(y - \bar{y})^2]$.)

Bagging: Why It Works

How does averaging the predictions affect the bias, variance and Bayes error?

$$y = \frac{1}{m} \sum_{i=1}^m y_i$$

- Bayes error: **unchanged**, since we have no control over it.
- Bias: **unchanged**, since the average prediction has the same expectation.

$$\bar{y} = \mathbb{E}[y] = \mathbb{E} \left[\frac{1}{m} \sum_{i=1}^m y_i \right] = \mathbb{E}[y_i]$$

- Variance: **decreased**, since we are averaging over independent samples.

$$\text{Var}[y] = \text{Var} \left[\frac{1}{m} \sum_{i=1}^m y_i \right] = \frac{1}{m^2} \text{Var} \left[\sum_{i=1}^m y_i \right] = \frac{1}{m^2} \sum_{i=1}^m \text{Var}[y_i] = \frac{1}{m} \text{Var}[y_i]$$

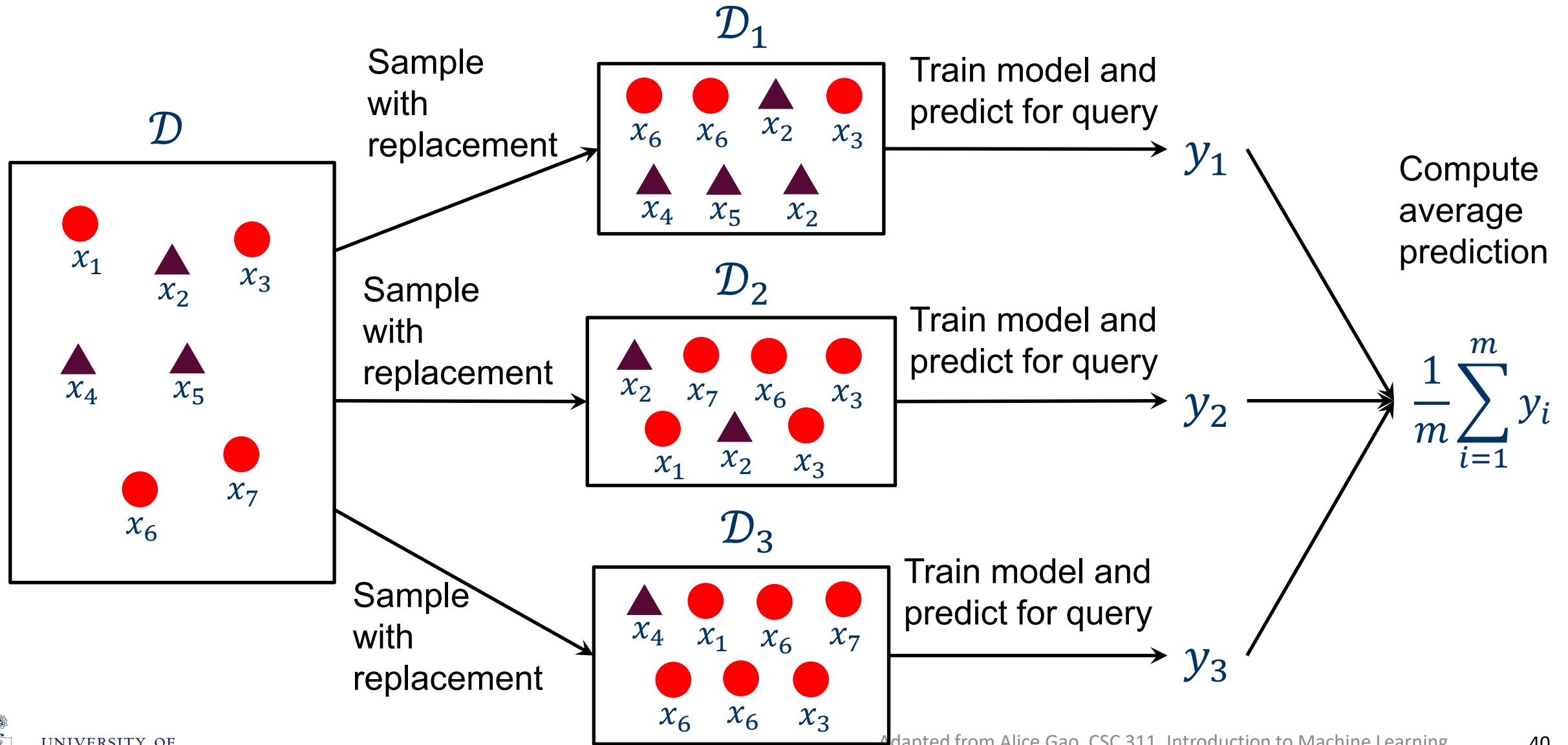
Bagging: How It Works

In practice, unrealistic to sample independent datasets.

Bagging (Bootstrap Aggregation)

1. Bootstrap sampling: creating multiple subsets of the original data set by randomly sampling with replacement.
2. Train: Build multiple models on these resampled datasets.
3. Aggregate: Average predictions across all models.

Bagging Visualized



Bagging: Aggregating Predictions

The classifiers may output probabilities $z_i \in [0, 1]$ or binary values $y_i \in \{0, 1\}$

In either case, we can produce a binary prediction $y_{bagged} \in \{0, 1\}$

by averaging the predictions and then applying a threshold.

$$y_{bagged} = \mathbb{I}\left(\frac{1}{m} \sum_{i=1}^m z_i > 0.5\right), \quad y_{bagged} = \mathbb{I}\left(\frac{1}{m} \sum_{i=1}^m y_i > 0.5\right)$$

($\mathbb{I}$ is the indicator function. $\mathbb{I}(True) = 1$, $\mathbb{I}(False) = 0$).

Bagging's Power: Reducing Variance Through Diversity

Bagging aims to **reduce variance** of predictions by averaging models.

If the models are trained on **highly correlated data**,

we won't achieve the full $\frac{1}{m}$ variance reduction with independent samples.

Solution: Sometimes we **intentionally introduce randomness**
in the training process to reduce correlation.

Bagging's **effectiveness** hinges on creating **diverse** models.

Random Forests (or Bagged Decision Trees)

1. Create multiple subsets of data by randomly sampling dataset with replacement.
2. Build a decision tree for each subset. At each split, **select a random subset of features** to determine the best split.

(Why? To reduce correlation between predictions)

3. Combine the predictions of all the trees.

Performance of Random Forests in Practice

- Probably the best black-box machine learning algorithm
- Often work well with no tuning whatsoever
- One of the most widely used algorithms in Kaggle competitions

Summary: Bagging

- Bagging reduces overfitting by averaging predictions.
- Commonly used in competition winners.
- Even if a single model is great, a small ensemble can enhance performance.

Limitations:

- Does not reduce bias when using squared error.
- Classifiers may still be correlated. Random forest introduces more randomness to reduce correlation.
- All members are weighed equally. May achieve better results by using a principled approach to weighted ensembling.